



Graphics, sound and software

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Spectravideo SV-318

The first widely available machine to be based on the proposed MSX hardware standard, the Spectravideo SV-318 is an attractive home computer system with some exceptional features. High resolution colour graphics, extensive sound effects and a powerful Basic interpreter combine to offer excellent value for money.

The SV-318 is a well-presented system, providing everything the new user needs to get started. The console is in white, impact-resistant plastic and measures 398 x 218 x 75mm (W x D x H at rear), sloping to 45mm at the front. There are 71 rubber pushbutton style keys arranged in a standard "qwerty" layout, with a row of special function and editing keys at the top of the keyboard.

Power is supplied by a separate metal-boxed unit with adequately long leads. The power connector is on the right side of the keyboard console, just behind the rocker-type power switch, and a small rectangular LED on the console top indicates that the computer is switched on.

Also on the top of the console is a slot

for software cartridges, normally covered by a hinged lid to protect the circuitry from prying fingers.

On the right side of the console are two connectors for external joysticks or games controllers. The pin-outs of these ports conform to the Atari arrangement so that many different types of joysticks can be used, apart from the three different controllers available from Spectravideo.

At the rear of the console is a cassette interface port, direct audio and video output (a 5-pin DIN socket) and a bus expansion connector. A VHF modulator is supplied so that SV-318 can be used with a standard television receiver on either channel 0 or channel 1. When the modulator is used sound signals are also

sent to the television and heard over the receiver's speaker. Using the direct video output to drive a monitor would require separate provision for sound.

Text is normally displayed in white on a light blue background and is stable and easily legible, using a direct video connection. The RF modulator is provided with a TV antenna switch box with rather long cables which seem to render it prone to interference.

A feature of the SV-318 is the built-in joystick. At the right of the keyboard is a circular "cursor control pad" pivoted in the centre. Four pushbuttons are arranged around the periphery, indicated by small depressions in the surface of the pad, while a hole in the centre takes a small plastic handle to create a non-proportional joystick.

We were also supplied with a Spectravideo SV-903 cassette drive, as standard cassette recorders cannot be used with the SV-318. The cassette drive derives power from the computer and comes equipped with a tape counter, "in use" LED indicator and an internal

◀ The SV-318 provides pushbutton keys and a built-in cursor control pad and joystick. Sprites and high resolution graphics in 16 colours allow excellent animated effects. ▶



home computer

microphone with a cut-out switch.

The microphone allows verbal comments to be added to taped programs as the SV-903 is a stereo recorder, with one channel reserved for programs and the other for voice. However the manual provides no details of how to use this feature.

Keyboard and video display

Extensive facilities are provided by the SV-318 keyboard, including upper and lower case letters, graphics, 10 programmable functions and editing keys.

Shift keys appear in the usual places, and the Caps Lock key is equipped with a rectangular LED indicator, a convenient feature, as the key itself does not lock down. Two sets of 26 graphics patterns are shown above the alphabetic keys and are accessed with the "Left Grph" and "Right Grph" keys which are both mounted to the left of the space-bar.

Five keys provide 10 programmable functions (with the use of the Shift key). Initially these default to frequently used

Basic keywords, such as CLOAD, COLOR and LIST, but are easily changed from Basic. The definition of each key is shown on the bottom line of the display screen, changing when the Shift key is pressed.

Although it is not mentioned in the manual, these prompts can be disabled with the statement:

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SCREEN,0
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and re-displayed with;

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SCREEN,1
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(Courtesy of the Spectravideo Computer Users Group of Tasmania).

Only two editing keys are provided, but the cursor control pad and provision for full screen editing allow very flexible manipulation of program lines on the display. Once a program is listed the cursor can be moved to any line and new characters typed over the old. The two editing keys, INS and DEL, operate to open up space within a line for new characters or to delete characters and close up those remaining.

One other key clears the screen and/or moves the cursor to the top left corner.

It also carried the label "COPY", which on some other computers is used to dump the contents of the screen to an attached graphics printer, but no information is available on this function. Information is also lacking on the use of the "Select" key, placed prominently on the extreme right of the keyboard.

The keyboard of the SV-318 is not recommended for typists. The flat keys are slightly larger than usual and well spaced, with limited travel. They also wobble uncomfortably, and must be struck at the exact centre to operate correctly. Hitting the edge of a key does not register as a key press, forcing the user to adopt a very careful typing style. Most users of the SV-318, however, will find the keyboard satisfactory. One-finger typing seems to be the norm, and the optional audible click as each key press is registered largely overcomes the peculiarities of the key movement. As such, the keyboard compares favourably with those of other computers in the same price range.

The Spectravideo provides three display modes — text in a 40 column by 24 line format, 64 x 48 resolution block graphics and 256 x 192 high resolution graphics. All modes allow a choice of 16 colours (including black and white) and both graphics modes allow the use of up to 32 "sprites".

"Sprites" are blocks of definable characters which are displayed independently of the remainder of the screen. Once defined a sprite is moved by simply updating its X and Y coordinates with the PUT SPRITE statement. The display controller then automatically places the sprite on the screen in a plane which takes priority over the text and background characters, so that a sprite, for example, can move over an existing display, leaving it undisturbed.

Sprites also have their own priority, so that sprite 0, for example, will be displayed on top of sprite 1, which is in turn displayed on top of sprite number 2.

Either 8 x 8 or 16 x 16 pixel sprites can be used and the ease of definition and movement, combined with the automatic priority feature make them ideal for fast moving games displays and other animated graphics.

Spectravideo Basic

The SV-318 comes with Microsoft Extended Basic in ROM, a language which fully supports the graphics, sound and programmable key functions and has other extensive capabilities which make it one of the most powerful versions of Basic available for home use.

A complete listing of the language is shown in Table 2, but some statements are especially noteworthy. AUTO enables the automatic generation of line

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numbers, while RENUM will renumber an existing program from a designated starting point, using a designated line number increment.

Once a program is written, the trace functions TRON AND TROFF are available to assist debugging. TRON, or "TRace ON" will enable tracing, printing each line number as the corresponding statement is executed and continuing until disabled with TROFF.

Graphics are supported with the PSET, PRESET, LINE, CIRCLE, PAINT and DRAW statements. The first two turn individual pixels on and off on the screen while LINE draws a line between two specified points. It takes parameters which specify the colour of the line and also enables a rectangle to be drawn with diagonal corners at the points specified. Simply adding an "F" to the statement will fill in the box to create a solid shape.

The CIRCLE statement draws a circle centred on a specified point, but it too takes parameters which specify the radius of the circle, the colour, the ratio of width to height (allowing ellipses to be drawn) and the starting and end points of the curve (allowing arcs and partial shapes to be created).

Once a line drawing has been created the PAINT statement can be used to fill an area with colour. The fill begins at a specified point and continues until a border of the designated colour is encountered.

The DRAW statement actually leads into a subset of Basic called a "Graphics Macro Language". DRAW takes a string of parameters which specify points to move to, the direction and length of lines drawn, and the scaling factor of a drawing. Additionally a separate string of drawing instructions can be included and called from the original string in the nature of a subroutine, allowing complex repetitive designs to be created quickly and easily.

GET and PUT are also interesting. GET allows a specified area of the screen to be stored in an array in memory, while PUT takes the information from the array and displays the shape on the screen. Together the two statements allow the use of "shape tables" for complex graphics and patterns, although they execute too slowly for use in animated displays. With sprites however, GET and PUT are not required for this use.

The sound effects and music capabilities of the SV-318 are extensive, and are well supported by the Basic interpreter. Sound is produced by a General Instruments AY-3-8910 chip, which provides three voices, each with a seven octave range, plus a noise source. The volume and wave shape of

each sound source can be separately controlled and the outputs combined in a programmable mixer by use of the SOUND statement.

In addition to SOUND for direct manipulation of the sound generator, the SV-318 Spectravideo Basic provides CLICK (for enabling and disabling audible feedback from the keyboard), BEEP and PLAY, a statement which introduce the "Music Macro Language".

PLAY takes a string expression which specified the notes to be played, octaves, flats and sharps, duration, tempo and volume, as in:
PLAY "CDEFGAB 02 CEF V15 GAB"

The wave shape and volume of the noise source can be specified, and as with the graphics macro language, substrings can be executed from within the main string.

Graphics and sound support are not the only exceptional features of Spectravideo Basic. Extensive string handling capabilities are provided, including hexadecimal, octal and binary string functions and an array of I/O and disk data formatting statements.

Four types of variables are supported, double precision, single precision, integer and string. Unless otherwise specified all variables will be treated as double precision types, which automatically performs calculations with 16 digit accuracy (better than many calculators).

The programmable function keys are supported by KEY(), which allows a string of up to 16 characters to be associated with each key, and KEY LIST, to display all current key definitions. Interrupts generated by a key press can re-direct the flow of program control using with the statement ON KEY () GOSUB. The internal timer, joystick

buttons and sprites also generate interrupts and are supported with their own statements.

In short, Spectravideo Basic is a comprehensive language with some intriguing new features, in keeping with the advanced hardware design of the Spectravideo computer.

The hardware

Spectravideo is so far the only company to market a computer based on the proposed MSX standard for 8-bit home computers. This standard arose from an agreement between Microsoft and 14 computer companies, including Spectravideo, NEC, Matsushita, Sony and Toshiba and other major Japanese companies.

The specifications of the standard so far cover only I/O port and major hardware functions and are intended to ensure that all MSX machines will run a common set of software. MSX machines are based on the Z80A microprocessor, the Texas Instruments 9918 video display processor (the PAL version is the 9929) and the General Instruments AY-3-8910 sound generator chip.

Input/output ports are provided by an Intel 8255 parallel I/O chip and an Atari type joystick interface and at least 32K of ROM for Microsoft's extended Basic are also specified.

In keeping with this standard the Spectravideo uses a Z80A processor running at 3.6MHz. The 40 x 24 text display, low and high resolution graphics modes and 32 sprites are provided by the 9929 VDP (the same chip is used in the TI-99/4A home computer), while the 8910 is responsible for the Spectravideo's extensive sound effects capabilities.

The standard SV-318 comes with 32K

Table 1: Spectravideo SV-318 specifications

Processor:	Z80A at 3.6MHz clock speed.
RAM:	32K (12.8K usable from Basic) expandable externally to 256K.
ROM:	32K expandable to 96K with cartridges.
Interfaces:	Cassette port, two games controller port, expansion connector.
Display:	40 x 24 line upper and lowercase text; 52 graphics characters; 64 x 48 and 256 x 192 pixel graphics in 16 colours; 32 sprites.
Sound:	Three voices, seven octave range, programmable noise source.
Software:	Microsoft Extended Basic in ROM, cassette and cartridge CP/M and Coleco adapter available.
Documentation:	Lacking detail for advanced users.

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of programmable memory, but of this, 16K is dedicated to the video display and another 3569 bytes is used for "house keeping" overhead, leaving just 12.8K available to the user. External cartridges are available in 16K and 64K sizes to expand this memory, and of course software ROM cartridges expand the Read Only Memory capability of the machine, although they have yet to appear in quantity.

For some reason the Spectravideo is not fully compatible with the MSX standard. The Spectravideo will need a special cartridge slot adapter to allow it to use MSX software cartridges when they appear but as there is no indication of when this will be the point is a minor one.

The other expansion possibilities of the SV-318 are also intriguing. Promised in brochures accompanying the computer is a games adapter which will allow the Spectravideo to use cartridges designed for the Coleco video games machines. Other possibilities include memory expansion to 256K, using an external expansion unit. This memory would be accessed by a bank switching scheme, currently supported by the Basic SWITCH statement.

A graphics tablet, disk drives and a printer are also on the way. To use these units, however, separate interfaces are required, and without an expansion unit only one can be used at a time. So far a disk drive controller, Centronics printer interface, 16K and 64K RAM modules and an RS232 interface are advertised.

The disk drive comes with a CP/M system disk, which (allowing for format incompatibilities) opens up another huge source of software for the SV-318. So far no details have been announced, and it is unclear exactly how CP/M disk programs can be transferred to the SV-318.

Software and documentation

For this review we were supplied with four programs on cassette tape. As expected, the games are very strong on graphics and sound effects and well up to the usual standard in the price range.

The "Basic Tutorial" was disappointing, consisting of a cassette of example programs and a booklet explaining what each does. No attempt was made to use the graphics, sound and audio from the cassette interface to assist instruction, so this series of programs does not do justice to the machine.

The home management system consists of two programs, one for keeping track of bonus coupons which are apparently offered with many products in the United States.

The second program is a recipe filing

Table 2: Spectravideo Basic statements

ABS, AND, ASC, ATN, AUTO, BEEP, BLOAD, BSAVE, CDBL, CHR\$, CINT, CIRCLE, CLEAR CLICK(ON, OFF), CLOAD, CLOSE, CLS, COLOR, COS, CONT, CSAVE, CSNG, CVD, CVI, CVS, DATA, DEF FN, DEF USR, DELETE, DIM, DRAW, EQV, ERASE, ERL, ERR, ERROR, EXP, FIX, FOR . . . STEP, FRE, GET, GOSUB, GOTO, HEX\$, IF . . . THEN . . . ELSE, IF . . . GOTO . . . ELSE, IMP, INKEY\$, INPUT\$, INSTR, INT, INTERVAL (ON/OFF/STOP), KEY(), KEY LIST, KEY ON/OFF/STOP, LEFT\$, LEN, LET, LINE, LIST, LLIST, LOAD, LOCATE, LOG, LPOS, MAXFILES, MDK\$, MERGE, MID\$, MKI\$, MKS\$, MOD, MOTOR ON/OFF, NEW, NOT, OCT\$, OPEN, ON ERROR GOSUB, ON . . . GOSUB, ON . . . GOTO, ON INTERVAL . . . GOSUB, ON KEY() GOSUB, ON STOP GOSUB, ON SPRITE GOSUB, ON STRIG GOSUB, OUT, PAINT, PEEK, PLAY, POINT, POKE, POS, PSET, PRESET, PUT, PUT SPRITE, REM, RENUM, RESTORE, RETURN, RIGHT\$, RND, RUN, SAVE, SGN, SIN, SCREEN, SOUND ON/OFF, SOUND (reg, exp), SQR, SPACE\$, SPC, SPRITE\$, SPRITE ON/OFF/STOP, STOP, STOP ON/OFF/STOP, STRING\$, STR\$, STRIG ON/OFF/STOP, SWAP, SWITCH, SWITCH STOP, TAB, TAN, TIME, TROFF, TRON, USR, VAL, VARPTR, VPEEK, VPOKE, WAIT, WIDTH, XOR.

system, a good example of the use of the cassette recorder for data storage.

Also announced is a spreadsheet package, home accounting programs, a word processing package, programmer's utilities and of course, more games. Coupled with the support for Coleco games cartridges (via an adapter) and a CP/M disk system, the SV-318 appears to be well supplied with software. The powerful Basic interpreter, sound and graphics support is also likely to encourage third party suppliers to provide software.

The most disappointing aspect of the SV-318 is the documentation. Two manuals are provided with the system, one a tutorial introduction to Basic and the other a reference guide. These are good as far as they go, but they don't go very far. The serious user is likely to quickly pose questions which the documentation does not answer, such as how to call machine language routines, use sprites directly (rather than through Basic) and details of programming the sound generator. In this respect, the data manuals on the 9918 VDP and the 8910 sound generator would be invaluable.

Certain Basic statements are either not mentioned or are dealt with incompletely by the manuals. There is a TIME function, for instance, that returns the value of a timer in the SV-318, but you must read newsletters from Spectravideo users groups to find out about it. The SWITCH function, which allows banks of RAM to be swapped in and out of the system memory is also unexplained, although it is mentioned.

This lack of information is most likely to affect the user who wishes to expand the system or write machine language software.

Pricing

As with all home computers, the Spectravideo SV-318 starts off inexpensively and ends up becoming an open invitation to spend money. The basic keyboard with 32K of RAM costs \$399, while the special purpose cassette recorder adds \$99 to this price.

From this point, pricing depends on how many peripherals are added. A Centronics interface, cable and dot matrix graphics printer are available as a package for \$699. A disk drive and controller together add \$798, but if you want to use both together you'll need either a "mini-expander" (two slots, for \$39) or the seven slot "super-expander" unit (\$249).

Given its initial low price, the SV-318 is good value for money. The wisdom of expanding the system is questionable however, as a system with a disk drive and printer costs will over \$1000 while the pushbutton keyboard remains unsuited for serious use. Users intending to create a fully expanded system might be better advised to look at the Spectravideo SV-328.

The SV-328 is the big brother of the machine reviewed here. It is software and hardware compatible with the SV-318 but comes with a full-stroke 87-key keyboard including a numeric pad and cursor control keys (there is no built-in joystick). Standard memory is 48K of ROM and 64K of RAM (with 29.2K usable). Like the SV-318, memory is expandable to 256K.

Our review machine was supplied by Computer Wave Pty Ltd, Lower Ground Floor, Grace Brothers Market Street store, or 325 George St, Sydney (02) 29 1631. The postal address is PO Box 268, GPO Sydney, NSW, 2001.